Course Final Test

Grade 6 Mathematics

Name: _____ Date: _____

Directions

This test has 48 questions in two test units of 24 questions. Work independently without a calculator. Read each question carefully. For multiple choice, select one answer; for multiple select, select every correct answer and no incorrect answer. Write exact numbers or expressions when requested, and label graph scales and units. Use the spaces provided for work.

This is original Math Coach practice modeled on public AASA item formats. It is not an official Arizona assessment.

Test Unit 1

Questions 1–24 | No calculator

01 / Constructed response

Use the standard algorithm: 39.5 + 8.2. Show your regrouping or division steps.

02 / Constructed response

Write 4^3 as repeated multiplication and evaluate.

03 / Constructed response

A tray has 5 red tiles and 5 blue tiles. Write the red-to-blue ratio in simplest form.

04 / Constructed response

A marker is 20 m below sea level. Write its signed elevation and explain what elevation 0 means.

05 / Constructed response

For 2x + 7y + 4, list terms, coefficients, and the constant.

06 / Multiple select

A class studies a collection of dogs. Select all statistical questions, and identify the expected variability.
A. How many centimeters are in 4 meters?
B. How does mass vary among the dogs?
C. What is 4 + 5?
D. What is the mass in kilograms of each of the dogs?

07 / Constructed response

Use long division: 300 ÷ 5. Show the algorithm and check.

08 / Constructed response

Solve x + 2 = 6. Check in the original equation.

09 / Multiple choice

Use the standard algorithm: 2.6 - 1.63. Show your regrouping or division steps. Select one answer, then justify it with a model or calculation.
A. 97/10
B. 97/100
C. 197/100
D. -3/100

10 / Multiple select

A recipe uses 3 cups oats for 2 cups seeds. Select all (oats, seeds) pairs that keep this ratio.
A. (6, 5)
B. (7, 4)
C. (6, 4)
D. (9, 6)

11 / Numeric entry and error analysis

Evaluate $3^2 + 4(5 - 3)$. Part A: enter the value. Part B: a student begins by adding 3^2 and 4; explain why that changes the expression.

12 / Multiple choice

Use the standard algorithm: 1.92 × 1.4. Show your regrouping or division steps. Select one answer, then justify it with a model or calculation.
A. 461/125
B. 211/125
C. 336/125
D. 672/25

13 / Multiple select

A triangle has base 14 cm and perpendicular height 5 cm. Select all expressions that give its area in cm².
A. 14 × 5
B. 7 × 5
C. (14 × 5) ÷ 2
D. (14 + 5) ÷ 2

14 / Expression entry and reasoning

A student buys x markers at $5 each and y folders at $2 each, then uses one $2 coupon. Part A: enter an expression for the total cost. Part B: explain why the coupon is subtracted only once.

15 / Multiple choice

Use the standard algorithm: 13.75 ÷ 2.5. Show your regrouping or division steps. Select one answer, then justify it with a model or calculation.
A. 55
B. 9/2
C. 13/2
D. 11/2

16 / Double number line and explanation

Draw a double number line for 3 liters of paint covering 3 m². Part A: align 0, 1, 2, and 7 batches. Part B: enter the area covered by 21 liters and explain why the scales can be different.

17 / Numeric entry and reasoning

Let x = 3/2 and y = 3. Evaluate 7x + y². Part A: enter the value. Part B: show substitution and the order of operations.

18 / Multiple choice

Draw a bar partitioned into 4 equal parts. Shade 2 parts. Circle groups of 1 parts. Use the model to find 1/2 ÷ 1/4. Select one answer, then justify it with a model or calculation.
A. 3
B. 2
C. 1
D. 20

19 / Multiple select

5 notebooks cost $25. Select all statements at the same price per notebook.
A. 3 notebooks cost $16.
B. Each notebook costs $125.
C. 3 notebooks cost $15.
D. Each notebook costs $5.

20 / Multiple select

Select all expressions equivalent to 2(x + 4).
A. 2x + 4
B. 2x + 8
C. 8 + 2x
D. 6x

21 / Multiple choice

Calculate 7/4 ÷ 3/5. Simplify. Select one answer, then justify it with a model or calculation.
A. 12/35
B. 23/12
C. 47/12
D. 35/12

22 / Two-part numeric entry

A 10 m by 10 m rectangle has a right-triangle corner removed with perpendicular legs 4 m and 6 m. Part A: enter the removed area. Part B: enter the remaining area and justify subtraction.

23 / Multiple select

Select all expressions equivalent to 5x + 6x + 5.
A. 16x
B. 11x + 6
C. (5 + 6)x + 5
D. 11x + 5

24 / Multiple choice

Calculate 5 ÷ 2 1/2. Simplify. Select one answer, then justify it with a model or calculation.
A. 2
B. 3
C. 1
D. 1/2

Test Unit 2

Questions 25–48 | No calculator

25 / Two-part numeric entry and reasoning

Store A sells 6 notebooks for $24. Store B sells 7 notebooks for $35. Part A: enter the unit price at each store. Part B: choose the better buy and explain why package totals alone are insufficient.

26 / Multiple select

Select all expressions equivalent to 6(x + 4).
A. 10x
B. 6x + 24
C. 6x + 4
D. 24 + 6x

27 / Multiple select

Select all true statements about 12 and 18. Explain why the greatest factor fits the kit situation.
A. 6 is the largest number of identical no-leftover kits from 12 pencils and 18 erasers.
B. 30 divides both counts.
C. 7 is their greatest common factor.
D. 6 divides both 12 and 18.

28 / Box plot and numeric entry

Six students recorded reading times (minutes): 5, 8, 8, 11, 11, 14. Part A: draw a box plot and label its five-number summary. Part B: enter the IQR.



Choose and label an equal-interval scale.

29 / Multiple select

Solve x - 5/2 = 2. Select every representation of the solution.
A. 18/4
B. 9/2
C. 11/2
D. 13/2

30 / Multiple select

A bicycle travels 40 km in 2 hours at constant speed. Select all true statements.
A. The speed is 42 km per hour.
B. In 2 hours it travels 40 km.
C. In 1 hour it travels 21 km.
D. The speed is 20 km per hour.

31 / Multiple select

Temperatures are -9 °C and -7 °C. Select all true comparisons.
A. -9 is warmer than -7.
B. -9 < -7
C. -7 lies to the right of -9 on a number line.
D. -9 > 0

32 / Multiple select

Solve 2 × x = 7. Select every representation of the solution.
A. 14/4
B. 9/2
C. 11/2
D. 7/2

33 / Multiple select

Use 1 feet = 12 inches. Select all statements equivalent to 7/2 feet.
A. 42 inches
B. 43 inches
C. 7/2 × 12 inches
D. 7/24 inches

34 / Multiple select

Consider -9 and 7. Select all true statements.
A. -9 is farther from zero than 7.
B. |-9| = -9
C. -9 > 7
D. |-9| = 9

35 / Multiple select

Solve x ÷ 3 = 13/6. Select every representation of the solution.
A. 26/4
B. 17/2
C. 13/2
D. 15/2

36 / Numeric entry and equation

A school club has 80 members; 15% take part in a project. Part A: enter the number of participants. Part B: write an equation and explain whether the answer is greater or less than the whole.

37 / Two-part numeric entry

A closed box is 6 cm by 2 cm by 3 cm. Part A: enter its surface area. Part B: the 6-by-2 top is removed; enter the remaining surface area and explain.

38 / Data display and explanation

Reading-time frequencies (minutes) are 0–4: 3, 5–9: 4, 10–14: 5, 15–19: 4. Part A: draw a histogram. Part B: enter the observation count and name every highest-frequency interval.

39 / Model and constructed response

A jug contains 2 L. Each full bottle needs 4/3 L. How many full bottles can be filled, and how much liquid remains?

40 / Multiple select

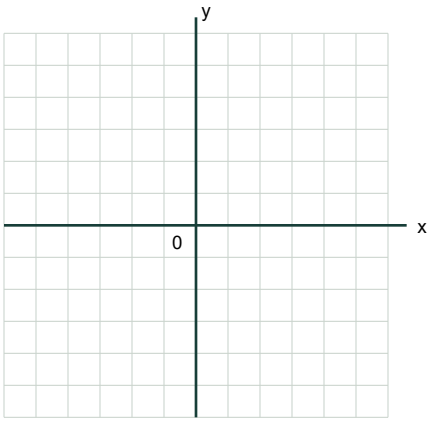
Select all values of x from the list that make x + 2 > 5 true. Explain why each rejected value fails.
A. 5
B. 3
C. 2
D. 4

41 / Multiple select

75 is 125% of an unknown quantity w. Select all equations that find w.
A. (125/100)w = 75
B. w = 75 + 125
C. w = 75 ÷ (125/100)
D. w = 75 × (125/100)

42 / Coordinate plotting and explanation

Plot P(-7, 4), Q(7, -4), and R(0, -4). Part A: name the quadrant of P and Q. Part B: explain why R is not in a quadrant.



Label the scale and units.

43 / Multiple select

A club needs at least 4 members. Select all true statements for its membership count n.

A. n ≥ 4

B. n > 4

C. 4 is an allowed membership count.

D. 3 is an allowed membership count.

44 / Numeric entry and model reasoning

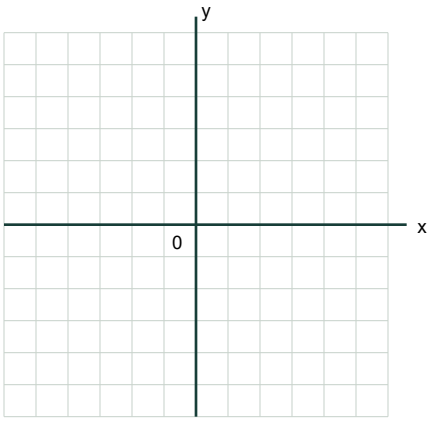
A prism measures 13/2 m by 5/2 m by 1 m. Part A: enter its volume. Part B: how many cubes with edge 1/2 m fill it? Explain why the cube count differs from the cubic-meter volume.

45 / Numeric entry and explanation

A quantity changes from a reference whole of 100 to a compared amount of 20. Part A: the compared amount is what percent of the reference whole? Part B: explain which number belongs in the denominator.

46 / Coordinate plotting and numeric entry

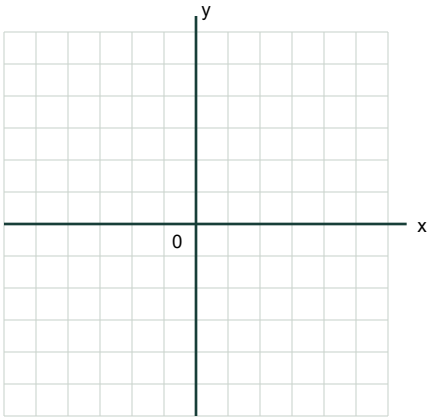
Plot A(-2, -3), B(3, -3), and C(-2, 3). Part A: enter AB. Part B: enter AC and explain how zero helps find each distance.



Label the scale and units.

47 / Table completion, equation, and graph

A machine uses 4 liters per hour. Part A: complete the (hours, liters) table: (0, __), (1, __), (2, __). Part B: choose the independent variable and write an equation for L in terms of h. Plot your pairs.



Label the scale and units.

48 / Statistical reasoning and explanation

Six students recorded reading times (minutes): 2, 8, 8, 14, 14, 28. Part A: describe center and spread using appropriate measures. Part B: describe shape, count, units, and any striking value; justify the chosen measures.